

Review

A Comprehensive Review of Position and Movement Visual Monitoring Systems with an Emphasis on AI Methods

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Featured Application

The outcomes of the provided analysis have significant practical application potential. All the authors are involved in the J-PET (Jagiellonian-PET TOMOGRAPHY) project, coordinated by the Faculty of Physics, Jagiellonian University. Grzegorz Filo is a team member; see the official project web page. The results are intended to be used to develop a concept, build models, and train and test a visual system for monitoring and supervising patient positioning during long-term whole-body CT scans.

Abstract

Recent systems for position and movement monitoring are increasingly enhanced by the integration of advanced AI techniques. Besides traditional analytical and algorithmic approaches, which are still applied in areas such as signal processing, sensor fusion, and kinematic modeling, there is a growing body of research that leverages AI-based methods to improve accuracy, robustness, and real-time decision-making capabilities. Artificial neural networks and deep learning methods are more and more often used for tasks such as predicting movement trajectories, detecting position anomalies, and approximating complex motion patterns. The main aim of this work is to provide the main contributions of the recent publications to the current state of the field. Key trends, challenges, and prospects for their future development are also highlighted. Initial statistical analysis was conducted based on responses to queries formulated for searching engines of leading online databases since 2006. Next, the retrieved articles from the last 6 years were subjected to a more detailed analysis. They were divided into thematic areas, including models for human pose estimation; systems for motion detection and tracking, with special attention to human movement; and, eventually, more specialized applications such as action recognition, autonomous driving, motion analysis, and surveillance. The architectures of the created models, the methods for parameter tuning or training, the input datasets used, and the result evaluation metrics were classified. Finally, some more general conclusions were drawn.

Keywords: object position; object movement; monitoring; tracking; artificial intelligence; neural networks; deep learning



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1. Introduction

Position monitoring and motion tracking play a key role in many fields, including engineering, mechanics, manufacturing, medicine, healthcare, rehabilitation, and more. Traditional methods for monitoring the position and movement of objects typically rely on

mechanical tracking systems, such as encoders and potentiometers, which directly measure angular or linear displacement. Optical tracking with markers, e.g., infrared-reflective markers observed by multiple fixed cameras, has been used in biomechanics and animation to reconstruct motion with high accuracy. Radio-based systems, including RFID and GPS, provide position estimates by measuring signal strength or travel time between transmitters and receivers. Inertial measurement units (IMUs), which combine accelerometers and gyroscopes, are components of traditional motion-tracking and estimating systems that use dead-reckoning techniques. Furthermore, ultrasonic and magnetic tracking systems are used in industrial and medical applications to determine object position based on propagation time or magnetic field distortion.

In turn, contemporary solutions increasingly rely on markerless, camera-based tracking. They replace earlier approaches that rely on sensors, enabling more accurate and scalable deployments. These systems can detect object position, track, and analyze motion by combining video input with AI-enhanced models that recognize patterns and predict trajectories in real time. The use of AI methods may improve precision, reduce time latency, and enable automated interpretation of the dynamic behavior of the investigated object. Modern position and movement monitoring systems offer evident advantages, including significantly improved accuracy through AI-enhanced pose estimation and the ability to operate without markers or specialized laboratory setups. They also enable real-time analysis in diverse environments, providing greater flexibility and applicability across industrial, medical, and human-machine interaction domains. However, these systems still face notable limitations, such as reduced robustness under poor lighting or dynamically changing backgrounds. Another weakness is their reliance on large datasets, which require lengthy training. Finally, many current solutions lack reliable uncertainty estimation, making them difficult to deploy when a guaranteed performance level is required.

Accordingly, this review presents the latest achievements and contemporary trends and development directions in AI-assisted systems for object position and movement monitoring. We wish to justify that integrating AI with modern visual, depth-based acquisition systems can significantly enhance capabilities such as high-fidelity position determination, pose reconstruction, motion prediction, and real-time trajectory tracking, even in the presence of disturbances and noise.

2. Materials and Methods

Preliminary studies led to the distinction of three main subject areas for further investigation. The first concerns human pose estimation, the second object detection and tracking, while the third relates to other, more specialized applications. Due to the credibility and high quality of the publications provided, ease of use, popularity among the academic community and the availability of refined, extensive search engines, it was decided to use the Scopus and Web of Science (WoS, Clarivate) databases. The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) reporting guidelines were obeyed. Eligibility criteria were defined before the search as follows:

- Inclusion criteria: (I1) Articles addressing position monitoring, movement monitoring, or position tracking of objects or humans. (I2) Articles employing at least one AI-based method (neural networks, deep learning, machine learning, or image processing). (I3) Journal articles, conference proceedings, and review papers indexed in Scopus or WoS. (I4) Publications available in English.
- Exclusion criteria: (E1) Articles primarily focused on closed-loop control systems (keywords: “control”, “position control”, “sliding mode control”, “controller”, “controllers”). (E2) Articles not available in full text. (E3) Duplicate records retrieved from both databases. (E4) Editorials, meeting abstracts, and grey literature.

The queries formulated for search engines based on the adopted eligibility and exclusion criteria are shown below.

%%% SCOPUS QUERY

TITLE-ABS-KEY (

("position monitoring" OR "movement monitoring" OR "position tracking")

AND

("neural networks" OR "machine learning" OR "AI" OR "neural network"

OR "image processing" OR "deep learning" OR "artificial intelligence")

AND NOT

("control" OR "position control" OR "sliding mode control"

OR "controller" OR "controllers"))

%%% WoS QUERY

TS = (

("position monitoring" OR "movement monitoring" OR "position tracking")

AND

("neural networks" OR "machine learning" OR "AI" OR "neural network"

OR "image processing" OR "deep learning" OR "artificial intelligence")

NOT

("control" OR "position control" OR "sliding mode control"

OR "controller" OR "controllers"))

The search results shown in Figure 1 were filtered by time to include statistics for publications limited to the last 20 years, since 2006. Remarkably, a continuously increased interest in the subject can be observed over the last 10–12 years, with the largest growth since 2023. Therefore, a more detailed analysis is presented in the latter part of this manuscript and applies to the most recent publications, mainly issued since 2019, with a few occasional references to older publications that make significant contributions.

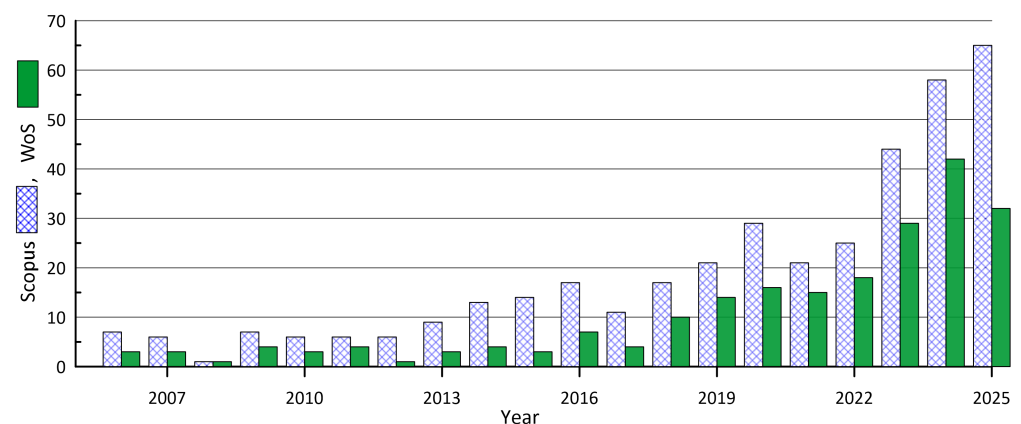


Figure 1. Number of publications on position and movement monitoring, registered in Scopus and WoS databases over the years.

The original query results were subjected to PRISMA analysis using the E1–E4 exclusion criteria. The obtained PRISMA flow diagram is presented in Figure 2.

After duplicate removal, the remaining records were manually checked for titles, abstracts, keywords, and, when necessary, the full text. This manual filtering step was used to verify whether each publication addressed at least one of the three review streams: human pose estimation, object detection and tracking, or specialized AI-assisted visual monitoring applications. Records were retained only when the study described an AI-based method, dataset, benchmark, or review directly relevant to position, pose, object

movement, detection, tracking, or activity analysis. Records were removed when their main contribution concerned closed-loop control, non-visual instrumentation, general image classification without position or movement analysis, unrelated medical or industrial monitoring, or methods outside the scope defined by the search criteria. In the final eligibility stage, publications older than 2019 were excluded unless they introduced a foundational architecture, benchmark, or evaluation concept needed to interpret recent work; this step corresponds to the “out-of-date” exclusion shown in Figure 2. The number of accepted publications from individual years is as follows: 2026 (31), 2025 (25), 2024 (9), 2023 (11), 2022 (5), 2021 (3), 2020 (9), 2019 (5), and older (3). Eight publications from the arXiv database (2020–2025) were also added, since they contain descriptions of key architectures and important performance test results. Finally, the analyzed set includes 109 publications. Subsequently, the main subject areas identified in most publications are shown in Figure 3.

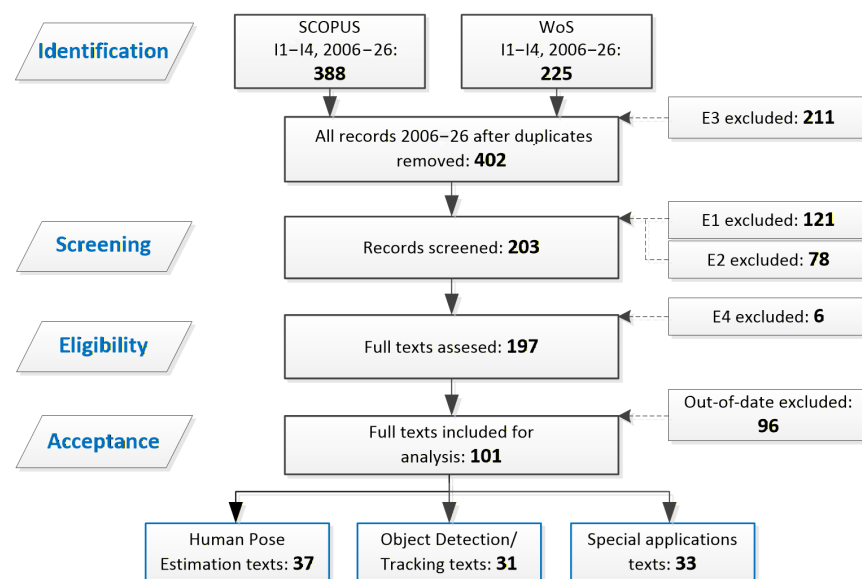


Figure 2. The obtained PRISMA flow diagram.

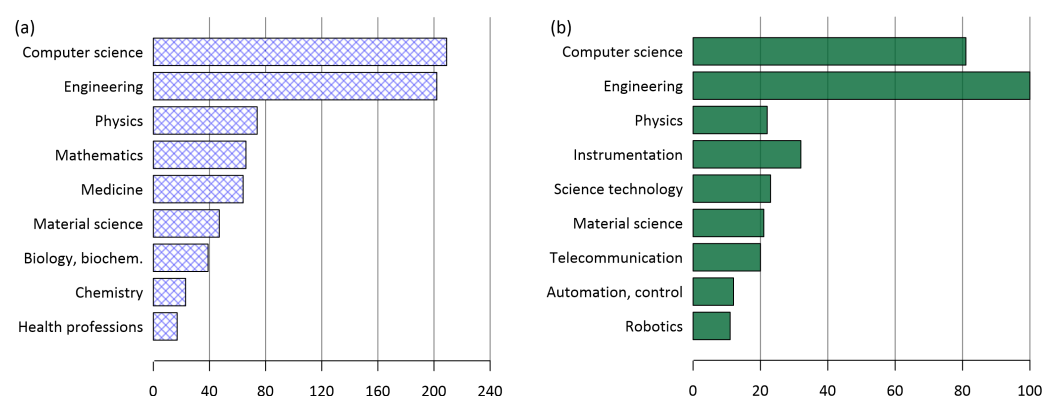


Figure 3. The most common subject areas of publications: (a) Scopus, (b) WoS.

It is worth noting that each publication can be assigned to multiple field names. The most frequent domains include computer science (Scopus: 209 and WoS: 91), engineering (202 and 108) and physics with astronomy (74 and 23), respectively. Other frequently mentioned fields include mathematics, medicine, material science, instrumentation, and chemistry. The following statistics concern the publication types shown in Figure 4.

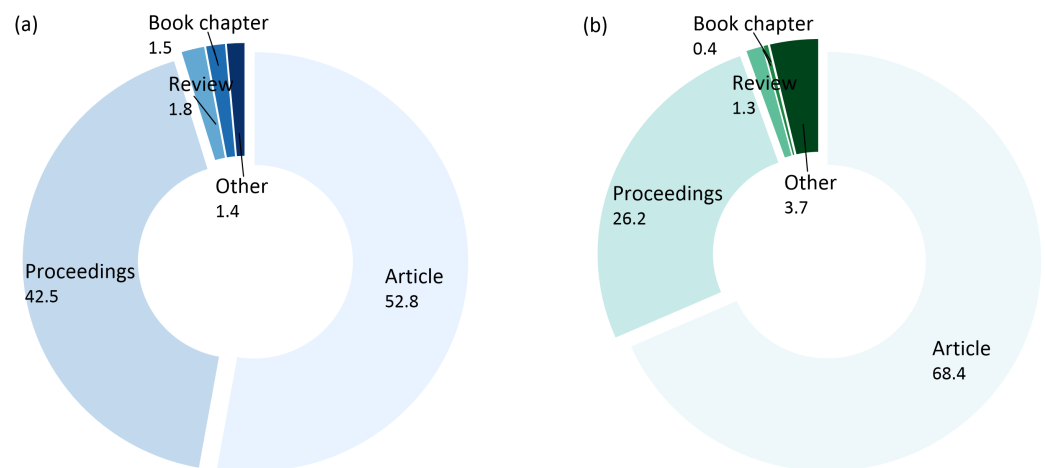


Figure 4. Types of publications: (a) Scopus, (b) WoS.

As for publication types, the dominant position is occupied by scientific articles, accounting for 52.8% in Scopus and 68.4% in WoS. The second place take conference proceedings, 42.5% and 26.2%, respectively. The remaining include book chapters, reviews, and other items, such as books, editorials, and meeting abstracts. Particularly, a small number of review studies can provide inspiration, motivation and encouragement for work in this area. The consecutive chart, presented in Figure 5, assigns publications to a country or region (territory). This assignment is based on the corresponding authors' country of affiliation. The results show that the People's Republic of China has the largest share of publications, followed by the USA, Germany, Italy and Japan. It is also worth noting India's high position in the Scopus database and a considerably larger share of publications registered in the WoS database from other countries.

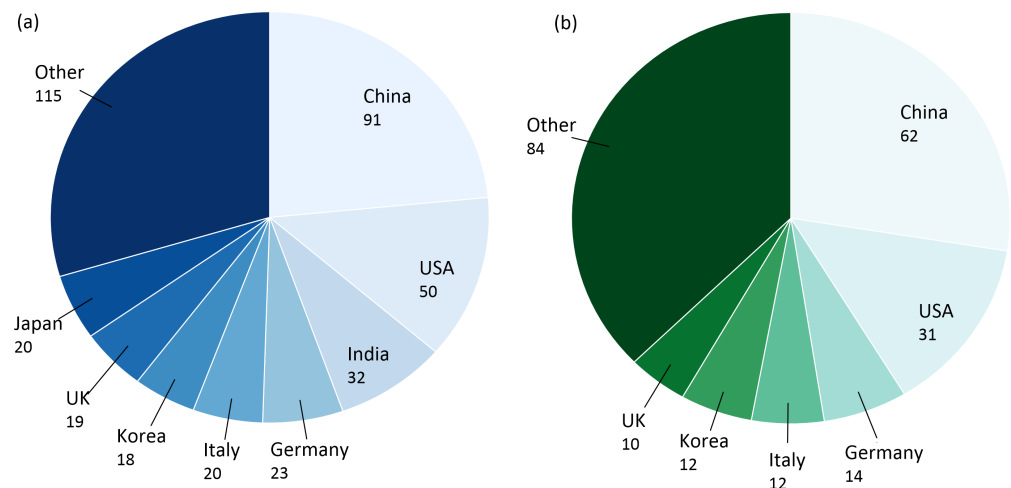


Figure 5. Publication assignment to country or region: (a) Scopus, (b) WoS.

The last summary concerns the number of documents available by open-access status. The “all access” level is assigned to 122 publications in the Scopus database and 100 in the Web of Science database within the searched subject area.

3. Conceptual Model-Selection Framework

Since position and movement monitoring systems differ strongly in input modality, output type, latency constraints, and deployment environment, the reviewed methods should not be selected only by their headline accuracy. To provide a unified author-driven synthesis across the broad range of topics covered in this review, Figure 6 presents a

decision-making framework that links practical design requirements with the families of models discussed in the following sections.

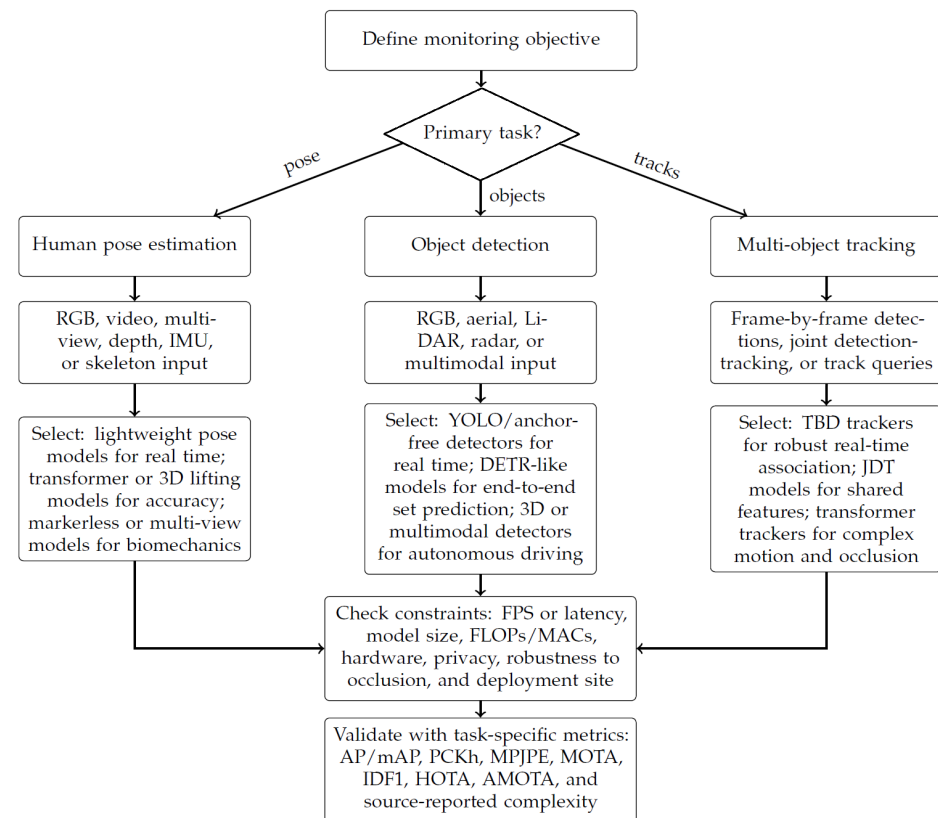


Figure 6. Author-driven decision-making flowchart for selecting model families in position and movement visual monitoring systems.

The flowchart is not intended to replace benchmark-specific validation; rather, it gives a structured first-pass guide for narrowing the candidate model class before comparing source-reported metrics such as AP, PCKh, MPJPE, MOTA, HOTA, FPS, and computational complexity. In the tracking branch, TBD denotes tracking-by-detection and JDT denotes joint detection and tracking.

4. Human Pose Estimation

Human pose estimation visual systems may include transformer-based methods, real-time methods, and markerless motion capturing. There are also available publications on survey methods, benchmark datasets, network architectures, and feedback systems. Tables 1 and 2 summarise representative publications discussed in the following subsections.

4.1. Survey Papers

Deep learning has quickly improved human pose estimation (HPE), leading to many surveys on 2D and 3D methods, different sensors, and practical applications. Zheng et al. [1] provide a review over 260 papers published up to 2022, organizing them by 2D or 3D approach, number of subjects, and input type. For 2D single-person estimation, they compare regression and heatmap methods. For multi-person cases, they discuss top-down and bottom-up pipelines. They also compare the results using benchmarks like MPII, COCO, PoseTrack, and Human3.6M. Dibenedetto et al. [2] review the development of methods from early CNN models to modern Transformers, focusing on uses in IoT

and smart environments. Their taxonomy for 3D estimation includes six types: direct prediction, 2D-to-3D lifting, SMPL-based volumetric models, and both monocular and multi-view image sequences. They also discuss privacy-preserving methods that use WiFi, RFID, and radar. Kappan et al. [3] deliver a broad survey from 2010 to 2025, organizing research by input type, number of subjects, and architecture. They review datasets such as LSP, MPII, COCO, Human3.6M, AthletePose3D, or WorldPose, and highlight issues like demographic bias and annotation limits that earlier reviews often miss. Samkari et al. [4] review about 100 papers on 2D HPE from 2014 to 2023, using strict selection and quality checks. They organize methods by input type (image or video) and number of subjects (single or multiple). The review covers datasets, loss functions, evaluation metrics, and backbone architectures. They find that CNNs are mostly used for spatial features, while RNNs are mainly used for modeling time in videos. Zhang and Shin [5] add a taxonomy based on task, output method (regression or heatmap), and architecture (top-down or bottom-up). Their survey shows heatmap-based methods are common and highlights progress from HRNet and transformer models like ViTPose and TokenPose. Ongoing challenges include weak cross-domain generalization and high computational needs.

Some reviews focus on 3D pose estimation. Salisu et al. [6] use the PRISMA protocol and organize 3D HPE methods by sensor type, such as RGB, IMU, depth, point cloud, and radio-frequency devices. Camera-based methods are divided into single-view single-person, single-view multi-person, and multi-view multi-person setups. The review notes that Human3.6M is the main benchmark and MPJPE is the most used metric, but current metrics do not fully capture temporal consistency or motion smoothness. El Kaid and Baïna [7] survey publications from 2021 to 2023 and introduce a new taxonomy based on input type and other factors. They distinguish between root-relative and camera-centric coordinate systems for multi-person estimation. Their results on Human3.6M show that transformer-based methods like MixSTE achieve a state-of-the-art MPJPE of 21.6 mm when using ground-truth 2D inputs. Neupane et al. [8] update the review to early 2024, grouping methods into eight scenarios based on input type, time structure, and number of subjects. They discuss semi-supervised, weakly supervised, and self-supervised learning, as well as deterministic, probabilistic, and diffusion-based approaches. Bataineh and Mohamed [9] focus on monocular estimation for ergonomic risk assessment (REBA) and note that few methods estimate REBA scores from single RGB images without depth sensors. Song et al. [10] assess both HPE and action recognition, finding that skeleton-based methods using OpenPose joints perform worse than video-based methods on Kinetics-400, mainly due to noisy pose estimates and limited temporal modeling. Tharatipyakul et al. [11] examine feedback systems for physical activity, reviewing 45 articles in exercise, sports, performing arts, and healthcare. They find OpenPose and MediaPipe are the most used libraries and point out the risk of incorrect feedback from imperfect pose estimates.

Iyer and Jeong [12] systematically review 35 studies on computer vision-based HPE for occupational safety, focusing on posture monitoring and ergonomic risk assessment. They discuss implementation challenges in industrial settings and analyze accuracy aspects of deployed HPE models, concluding that further real-world validation and practical integration remain necessary.

4.2. Transformer-Based Methods

Transformer architectures have greatly improved HPE in both 2D and 3D tasks (Table 1). Xu et al. [13] introduce ViTPose and ViTPose++, which use a standard ViT encoder for patch features and a lightweight decoder for keypoint heatmaps. The backbone comes in several sizes, from ViT-S to ViTAE-G, with the largest model reaching 80.9 AP on the MS COCO test-dev set. The authors show that freezing the multi-head self-attention

layers during fine-tuning only slightly reduces accuracy, so the FFN layers are most important for learning the task. ViTPose++ supports multi-task learning across human, whole-body, and animal datasets by splitting FFN layers into shared and task-specific parts, achieving top results on MS COCO, OCHuman, MPII, COCO-Wholebody, AP-10K, and APT-36K. To address the coordinate quantization problem in heatmap-based methods, Liu and Zheng [14] introduce VitCC, a Vision Transformer Coordinate Classification pipeline that treats keypoint localization as a pixel-based classification task. Their architecture uses an Effective Sparse Modeling Transformer block, which includes ESP Attention and an EM Feed-Forward Network. This design allows for parallel multi-scale token aggregation with different down-sampling rates in one self-attention layer. Tests on COCO and MPII show that VitCC matches RTMPose in speed and performance, achieving 75.8 AP on COCO and 88.8 PCKh on MPII with the CSPNeXt-m backbone. MotionAGFormer-B, a base variant, achieves a P1 error of 38.4 mm on Human3.6M, using about one quarter of the parameters and three times fewer multiply-accumulate operations than the previous top model, MotionBERT. The large variant reaches 16.2 mm MPJPE and 85.3% AUC on MPI-INF-3DHP. For multi-view 3D pose estimation, Liao et al. [15] proposes MVGFormer, a hybrid model that switches between a learnable Appearance Module and a non-learnable Geometry Module. They use differentiable algebraic triangulation on refined 2D poses to update 3D estimates. This step-by-step refinement avoids quantization errors seen in voxel-based methods like VoxelPose and prevents camera overfitting found in methods like MvP. Tests on CMU Panoptic, Shelf, and Campus benchmarks show that MVGFormer outperforms fully learned models in new settings and matches their performance in familiar ones. Lu and Min [16] introduce AiPE, which splits input channels into two groups and processes them in parallel with axis-specific self-attention, reducing computational cost compared to full spatio-temporal attention. On Human3.6M with 243 input frames, STCFormer-L achieves a Protocol 1 error of 40.5 mm and Protocol 2 error of 31.8 mm, using 18.91 million parameters compared to 33.61 million for MixSTE. In Table 1, NR indicates that the corresponding speed or the complexity indicator was not reported in the analyzed source, while AP25 denotes average precision at a 25 mm threshold. The values should be read as source-reported indicators rather than as a direct leaderboard, because datasets, input resolutions, and evaluation protocols differ across studies.

Recent works continue to refine transformer-based 3D pose pipelines, often combining attention with explicit structural priors. Zhao et al. [17] propose the Clip-Cross Hierarchical Refinement Network (CCHRN) for monocular video, combining a Transformer-based Clip-Cross Refinement Module (CCRM) with a hierarchical regressor that embeds SMPL-derived kinematic priors; on 3DPW and Human3.6M they report PA-MPJPE of 49.3 mm and 44.3 mm, respectively. Duan et al. [18] introduce MixTGFormer, a dual-stream spatio-temporal GCN-Transformer that treats joints as tokens in a spatial branch and frames as tokens in a temporal branch, reporting P1 errors of 37.6 mm (Human3.6M) and 16.5 mm (MPI-INF-3DHP). Yang et al. [19] present DDCEFormer with enhanced multi-head attention and cross-enhanced fusion of spatial and temporal features, achieving MPJPE/P-MPJPE of 39.1/30.8 mm on Human3.6M and PCK/AUC/MPJPE of 99.3%/88.8%/13.4 mm on MPI-INF-3DHP. To integrate anatomical structure into attention more explicitly, Wang et al. [20] propose MuRE, which augments self-attention with an Adaptive Multi-Relationship Attention (AMuRA) mechanism that incorporates both a joint self-similarity matrix and a skeleton adjacency prior. Chen et al. [21] propose SFTFormer, a structure-aware dual-stream framework that combines graph-based and attention-based modeling with dynamic topology-aware positional encoding and structure-enhanced spatio-temporal attention, reporting state-of-the-art performance on Human3.6M and MPI-INF-3DHP particularly for cross-limb interactions and fast motions while maintaining good efficiency. Zhou et al. [22]

introduce HSG-GCNFormer, which applies Discrete Cosine Transform (DCT) and spectral gating before fusing a spatial GCN stream with a spatio-temporal interlaced attention stream, reporting competitive performance on Human3.6M and MPI-INF-3DHP.

Table 1. Transformer-based publications discussed in the human pose estimation section.

Method	Input	Datasets	Reported Accuracy	Efficiency/Complexity
ViTPose++ [13]	RGB image	COCO, MPII, OCHuman, COCO-WholeBody	79.1 AP (COCO val); 80.9 AP (test-dev, ViTPose-G)	632 M params; 122.9 GFLOPs (ViTPose-H); no extra inference cost vs. ViTPose
VitCC [14]	RGB image	COCO, MPII	75.8 AP; 88.8 PCKh	12.72 M params; 2.01 GFLOPs; 71.219/158.729 FPS CPU/GPU
MVGFormer [15]	Multi-view RGB	CMU Panoptic, Shelf, Campus	92.3 AP25/16.0 MPJPE; 98.0/96.7 PCP	Geometry-guided refinement; computation NR
AiPE [16]	RGB image	COCO2017	86.3/68.9 AP at IoU 0.50/0.75	681.2 M params; 117.3 GFLOPs
CCHRN [17]	RGB video	3DPW, Human3.6M, MPI-INF-3DHP	49.3 PA-MPJPE/69.9 MPJPE (3DPW); 44.3 PA-MPJPE (Human3.6M)	68.25 M params; 4.70 GFLOPs; 79.0 ms/frame
MixTGFormer [18]	2D keypoints	Human3.6M, MPI-INF-3DHP	37.6/31.8 mm P1/P2; 16.5 mm MPJPE (MPI-INF-3DHP)	15.7 M params; 63.4 GMACs; 9.5 FPS
DDCEFormer [19]	2D keypoints	Human3.6M, MPI-INF-3DHP	39.1/30.8 mm MPJPE/P-MPJPE; 99.3 PCK/88.8 AUC/13.4 MPJPE	53.8 M params; 222 GFLOPs; 4858 FPS
MuRE [20]	2D keypoints	MPII, Human3.6M, MPI-INF-3DHP, Shelf	48.5/38.2 mm MPJPE/P-MPJPE; 86.1 PCK/54.2 AUC	5.54 M params; 94.8 MMACs
SFTFormer-B [21]	2D keypoints	Human3.6M, MPI-INF-3DHP	37.2/31.2 mm MPJPE/P-MPJPE; 99.2 PCK/90.6 AUC/14.4 MPJPE	29.5 M params; 52.8 GMACs; 217 M MACs/frame
HSG-GCNFormer [22]	2D keypoints	Human3.6M, MPI-INF-3DHP	42.8 MPJPE (Human3.6M); 23.9 MPJPE/98.3 PCK/81.6 AUC	4.76 M params; 0.730–6.832 GFLOPs

4.3. Real-Time and Efficient Methods

Achieving high accuracy with limited computing power is a major challenge in practical pose estimation (Table 2). Bazarevsky et al. [23] developed BlazePose, a convolutional neural network for single-person, on-device use. The model uses an encoder–decoder heatmap branch during training and a regression encoder for inference, which removes the need for heavy heatmap outputs and lowers latency. A face-detector-based tracking system also helps compress the model by only estimating poses when the head is visible. Lite-HRNet-18, with just 1.1 million parameters and 0.20 GFLOPs, achieves 64.8% AP. On the COCO val set, this is more than 9 AP points better than the Small HRNet baseline and uses much less computing power. Building on this, Li et al. [24] introduce Dite-HRNet, which adds a dynamic multi-scale context (DMC) block and a Dynamic Global Context (DGC) block to the multi-resolution backbone. The DMC block uses depth-wise convolutions with different kernel sizes in parallel and combines them using attention weights based on the input, while the DGC block performs resolution-adaptive context pooling. Dite-HRNet-30 achieves 71.5% AP on COCO val2017 at 384×288 input, beating Lite-HRNet-30 by 1.1 AP points with similar parameter and FLOPs counts. Jiang et al. [25] propose RTMPose, a real-time framework that uses a SimCC-based coordinate classification head, turning keypoint localization into separate classifications for horizontal and vertical axes. The CSPNeXt backbone balances speed and accuracy well compared to other backbones. RTMPose-m achieves 75.8% AP on COCO val2017 at over 90 FPS on the Intel i7-11700 CPU and over 430 FPS on the NVIDIA GTX 1660 Ti GPU. RTMPose-s gets 72.2% AP at over 70 FPS on a Snapdragon 865 mobile chip. Zhang et al. [26] present an Ultra-lightweight End-to-end

Pose Distillation Network (UEPDN), which combines a multi-stage expert network with a re-parameterized subnetwork and online distillation, where expert and student networks train together. Tests on the LSP dataset show that pose distillation increases Stage 1 accuracy from 89.5 to 90.7 PCK@0.2, and combining distillation with re-parameterization gives 87.5 PCK@0.2 with only 2.75 million parameters and 6.2 GFLOPs. In Table 2, the values are reported as given in the original papers. FPS and latency figures depend strongly on the tested hardware and runtime backend, while GFLOPs, MFLOPs, and MACs are not strictly interchangeable but are retained to preserve the source-reported complexity indicators.

Table 2. Real-time and efficient publications discussed in the human pose estimation section.

Method	Input	Datasets	Reported Accuracy	Speed/Complexity
BlazePose Full [23]	RGB image	AR, Yoga	84.1/84.5 PCK@0.2	102 FPS; 3.5M params; 6.9 MFLOPs
BlazePose Lite [23]	RGB image	AR, Yoga	79.6/77.6 PCK@0.2	312 FPS; 1.3M params; 2.7 MFLOPs
DITE-HRNet-18 [24]	RGB image	COCO val2017, MPII	69.0 AP; 87.0 PCKh	1.1M params; 0.4/0.2 GFLOPs
DITE-HRNet-30 [24]	RGB image	COCO val2017, MPII	71.5 AP; 87.6 PCKh	1.8M params; 0.7/0.4 GFLOPs
RTMPose-m [25]	RGB image	COCO val2017	75.3 AP; 73.2 pipeline AP	1.93 GFLOPs; 11.060 ms CPU (~90 FPS); 2.288 ms GPU (~437 FPS); 32.122 ms mobile
RTMPose-s [25]	RGB image	COCO val2017	71.2 AP; 68.5 pipeline AP	0.68 GFLOPs; 4.481 ms CPU (~223 FPS); 1.392 ms GPU (~718 FPS); 21.683 ms mobile
UEPDN-R1 [26]	RGB image	MPII, LSP	89.3 PCKh, 87.5 PCK@0.2	2.75M params; 6.2 GFLOPs; 5.3 FPS
AGMamba-L [27]	2D keypoints	Human3.6M, MPI-INF-3DHP	38.2/30.5 mm MPJPE/P-MPJPE; 98.2 PCK	4.5M params; 34.6 GMACs

Efficiency for long temporal windows is also addressed by Ma et al. [27], who propose AGMamba for continuous 2D-to-3D lifting. Their approach replaces the quadratic-cost self-attention bottleneck with a selective state-space model (Mamba) of linear complexity and combines it with an attention-enhanced GCN to jointly model spatial skeleton structure and temporal motion, reporting state-of-the-art performance on Human3.6M and MPI-INF-3DHP at low computational cost.

4.4. 3D Pose Estimation and Markerless Motion Capture

A comprehensive mini-review by Edriss et al. [28] examines technologies ranging from Microsoft Kinect and StereoLabs ZED cameras to open-source software libraries such as OpenPose, MediaPipe, AlphaPose, and DensePose, emphasising that the integration of vision sensors with AI-based 2D-to-3D lifting methods represents the most promising direction for future kinematic and biomechanical research in sports and clinical applications. Beyond conventional camera pipelines, Liu et al. [29] propose an image-free 3D pose estimation approach based on single-pixel imaging (SPI) and the MMPose architecture, extracting 3D pose information directly from low-sampling-rate measurement signals without full image reconstruction. On Human3.6M they report accurate predictions at sampling rates as low as 2.56×10^{-4} and validate real-world operation at distances up to 30 m under natural lighting, aiming to eliminate privacy leakage at the source and reduce transmission and storage overhead. Nakano et al. [30] developed a multi-camera system combining OpenPose-based 2D keypoint detection with Direct Linear Transformation (DLT) to reconstruct 3D joint coordinates in a global reference frame. Their evaluation across walking, countermovement jumping, and ball-throwing tasks demonstrated that approximately 80% of computed Mean Absolute Errors fell below 30 mm when compared against a 16-camera optical marker-based gold standard, with larger errors attributed to frame-level tracking failures in OpenPose, including misidentification of objects as body

segments or left–right limb swapping. Pavllo et al. [31] proposed a fully convolutional architecture exploiting dilated temporal convolutions over sequences of predicted 2D keypoints. On the Human3.6M benchmark, the model achieved a mean per-joint position error (MPJPE) of 46.8 mm, improving upon the previous best result by approximately 6.0 mm. The authors further introduced a semi-supervised back-projection scheme in which 3D pose estimates derived from unlabeled video are reprojected to 2D and compared with the original keypoint detections, enabling the exploitation of large unlabeled corpora. Even with as few as 5000 labeled frames, the semi-supervised approach yielded improvements of up to 15.0 mm MPJPE over the supervised baseline. Dill et al. [32] investigated the feasibility of 3D pose reconstruction using stereo camera fusion with the MediaPipe Pose framework, achieving a median RMSE of 30.1 mm across 81 recordings from nine subjects for squat exercise analysis, significantly outperforming monocular 3D MediaPipe estimation (median RMSE of 56.3 mm, $p < 10^{-6}$). For point cloud inputs, Chen et al. [33] propose SARL, combining a Geometric Guide Attention module that establishes region-wise local coordinate systems with a Frequency-Enhanced Point Transformer using DCT-based modulation to enhance low-frequency human configuration signals while suppressing high-frequency noise. They report consistent gains on the Sloper4D and Human-M3 datasets. Vun et al. [34] evaluated 17 studies on vision-based motion capture for gait analysis in neurodegenerative diseases, distinguishing between gait measurement systems and gait classification systems. Among AI-assisted approaches, Support Vector Machines were the most widely employed classifiers, and classification accuracies ranged from 62.1% to 97.0% across studies covering Parkinson’s disease, Alzheimer’s disease, dementia, and Multiple Sclerosis. Beyond gait-specific applications, recent work has increasingly addressed pose classification for structured activity recognition in image-based settings. Saber et al. [35] proposed a lightweight attention-based architecture combining a frozen Swin Transformer backbone with a cascade of Enhanced Spatial Attention (SPA), Context-Aware Channel Attention (CCAM), and Dual-Weighted Cross-Attention (DWCA) modules for the classification of yoga poses and general human actions. In turn, Shan et al. [36] proposed a Diffusion-based 3D Pose estimation method (D3DP) combined with a Joint-wise reProjection-based Multi-hypothesis Aggregation scheme (JPMA), leveraging Denoising Diffusion Probabilistic Models to generate multiple plausible 3D pose hypotheses from a single 2D observation. A key architectural advantage of D3DP is its compatibility with existing deterministic 3D pose estimators, demonstrated by employing MixSTE as the denoiser backbone with only minor modifications.

4.5. Benchmark Datasets, Network Architectures, and Feedback Systems

Liu et al. [37] introduced the NTU RGB+D 120 dataset, a large-scale RGB+D benchmark comprising 114,480 video samples collected from 106 subjects performing 120 action categories. The dataset was captured using Microsoft Kinect v2 sensors across 155 distinct camera viewpoints, 32 collection setups, and 96 different background environments, providing four data modalities: depth maps, 3D skeletal joint data (25 body joints), RGB video, and infrared sequences. Experimental evaluation demonstrated that combining RGB, depth, and skeleton modalities yields the highest recognition accuracy (64.0% cross-subject, 66.1% cross-setup), confirming the complementary nature of different data streams. The authors also proposed the Action-Part Semantic Relevance-aware (APSR) framework for one-shot 3D action recognition, leveraging Word2Vec embeddings to estimate semantic relevance scores between action and body part names, achieving 45.3% accuracy on one-shot recognition across 20 novel action classes. At the network architecture level, Ci et al. [38] introduced a Locally Connected Network (LCN) as a generic reformulation of the Graph Convolutional Network (GCN) operator for 3D human pose estimation from 2D joint detections. By factor-

ing the Laplacian matrix into a structure matrix and a weight matrix, the authors revealed that GCN suffers from an inherent weight-sharing scheme that restricts its representational capacity for pose regression tasks. The LCN retains sparse joint connectivity consistent with human anatomical structure while removing the weight-sharing constraint, achieving a mean per-joint position error (MPJPE) of 52.7 mm under Protocol #1 on Human3.6M. Importantly, LCN requires substantially fewer training samples to converge—approximately 2000 examples compared to over 5000 required by the FCN baseline—and generalizes more robustly to unseen action categories and cross-dataset scenarios. Ullah et al. [39] presented a computer vision-based action scoring system integrating the MediaPipe Pose framework with Dynamic Time Warping (DTW) and Normalized Cross-Correlation (NCC) for quantitative evaluation of rehabilitation exercises. The system extracts 33 anatomical landmarks per frame, derives joint angles using the cosine rule as the primary kinematic feature, and employs DTW to temporally align user and reference motion sequences while NCC measures trajectory similarity. The system achieved an average keypoint detection deviation below 10% relative to manually annotated ground truth, DTW alignment accuracy above 90%, and NCC similarity scores above 0.85 across the tested exercises. Comparative evaluation against RepNet demonstrated that the proposed angular approach outperformed RepNet in both accuracy and computational efficiency, particularly for exercises involving lower limb movements where RepNet frequently failed to detect any repetitions. Tharatipyakul et al. [11] systematically characterized deep learning-based HPE systems for physical movement feedback, identifying a three-module pipeline common to feedback systems: a pose estimation module, a movement assessment module, and an augmented feedback presentation module. The review identified open-source libraries—particularly OpenPose and MediaPipe—as dominant in practical adoption, with joint angle measurement as the most prevalent spatial feature and DTW as the exclusive temporal alignment method across reviewed papers.

5. Object Movement Detection and Tracking

Object detection and multi-object tracking (MOT) form the perceptual backbone of AI-based human monitoring systems. This section surveys the principal architectural families for 2D and 3D object detection, the dominant MOT paradigms and their data-association strategies, the key public benchmarks, and outstanding open challenges. Table 3 summarizes the representative methods discussed below.

5.1. Object Detection Methods

Deep learning detectors are commonly grouped into four families—two-stage, one-stage, anchor-free, and transformer-based—according to how region hypotheses are generated and subsequently refined [40]. Rather than progressing uniformly in accuracy, these families represent genuinely different engineering trade-offs: two-stage pipelines prioritise precision at the cost of throughput, while one-stage and anchor-free designs favor latency, and transformer architectures sacrifice convergence speed for architectural simplicity.

5.1.1. Two-Stage Detection Methods

Two-stage detectors decompose the detection problem into a region-proposal step followed by per-region classification and bounding-box regression. The R-CNN lineage pioneered this paradigm: candidate regions are extracted, classified by a convolutional classifier, and refined by a regression head. Faster R-CNN replaced the external selective-search procedure with an internal Region Proposal Network (RPN) that shares convolutional features with the detection head, enabling near-real-time operation while maintaining high mean average precision (mAP) [41]. A further structural contribution was the Feature

Pyramid Network (FPN), which connects high-resolution but semantically shallow features with semantically rich high-level ones through top-down lateral connections, substantially improving multi-scale detection [42].

The two-stage design has proven particularly well-suited to scenarios where detection quality matters more than frame rate—notably in stereo-based autonomous driving, where Yebes et al. [43] showed that 3D-aware features derived from stereo imagery yield substantial gains over monocular pipelines on the KITTI urban benchmark. That said, the accuracy advantage carries a real cost: inference latency in two-stage networks remains too high for many real-time monitoring deployments, and this bottleneck has driven the field progressively toward one-stage alternatives [40]. An extended two-stage pipeline combining automated image annotation with Inception v2-based Faster R-CNN (AIA-IFRCNN) was proposed by Vijiyakumar et al. [44]. The Discriminative Correlation Filter with Channel and Spatial Reliability Tracking (DCF-CSRT) model first annotates bounding boxes automatically, feeding structured labels to a Faster R-CNN detector; an Inception v2 feature extractor then produces class probabilities via a softmax layer. Evaluated on three heterogeneous benchmarks (Bird, UCSDped2, and Underwater), AIA-IFRCNN achieved peak detection accuracy of 95.62% and Centre Location Errors as low as 4.16 pixels, with Overlap Rates above 0.90 across all three datasets [44]. The integration of a correlation-filter tracker into the annotation loop illustrates how SOT components can serve as data-preparation tools for two-stage detectors, rather than only as runtime inference modules.

5.1.2. One-Stage Detection Methods

One-stage detectors predict class labels and bounding-box coordinates in a single forward pass over a fixed anchor grid, trading a portion of accuracy for substantially higher throughput. SSD (Single-Shot MultiBox Detector) set an early precedent by attaching multi-scale detection heads to different backbone layers, allowing objects of varying sizes to be captured without a proposal stage [45].

The YOLO (You Only Look Once) family carried this design principle furthest. Beginning with YOLOv1's direct grid regression at 45 FPS on PASCAL VOC, each successive generation refined the architecture in a distinct way [45]. YOLOv4 combined Cross-Stage Partial connections (CSP), Path Aggregation Network (PAN), and Mish activation to push the speed–accuracy frontier on MS COCO [42]. YOLOv7 extended this with trainable bag-of-freebies and ELAN-based feature aggregation, setting a new state-of-the-art speed–accuracy trade-off at the time of its publication [46].

YOLOv8 marked a design inflection point by migrating to an anchor-free detection head and adopting a C2f backbone module, achieving the highest accuracy in the series up to that point on COCO test-dev [47]. YOLO11 continues the line with a refined C3k2 block and depth-separable convolutions that improve the parameter-to-accuracy ratio compared with YOLOv8. Because YOLO11 is a very recent release (late 2024), peer-reviewed coverage is limited: the most thorough architectural comparisons available are the arXiv survey by Hidayatullah et al. [48] and the concurrent overview by Khanam and Hussain [49], cited here specifically for their detailed ablations spanning YOLOv8 through YOLO11. A comprehensive evolutionary account of the YOLO lineage from v1 to v8, covering improvements in backbone depth, anchor design, and data-augmentation strategy, is provided by [50]. The YOLO11-specific contribution of Alif [41] extends coverage to the latest generation in the context of intelligent transportation systems, reporting improved parameter-to-accuracy ratios relative to YOLOv8 on standard detection benchmarks.

5.1.3. Anchor-Free Detection Methods

Anchor-free detectors remove the engineering burden of predefined anchor scales and aspect ratios, localising objects instead through keypoint heatmaps or dense per-pixel predictions. CenterNet represents each object as a single centre point on a heatmap, with width, height, and optional attributes predicted as offsets from that point—a formulation that substantially simplifies the pipeline without sacrificing competitive accuracy on pedestrian and vehicle benchmarks [51]. This keypoint-centric design proved influential beyond detection: it became the foundation of the CenterTrack joint detector-tracker (Section 5.2.2).

The anchor-free philosophy naturally handles the extreme scale variation encountered in aerial and UAV imagery, a domain where standard anchor grids fail because objects may span only a handful of pixels. Slicing-Aided Hyper Inference (SAHI) addresses this by tiling high-resolution images into overlapping sub-patches, running inference on each slice, and merging results via Non-Maximum Suppression. Applied to RT-DETR-X on the VisDrone-DET dataset, SAHI raised average precision by 3.3 percentage points, though throughput fell from roughly 28 FPS to just over 1 FPS—a trade-off that constrains its use to offline or near-real-time pipelines [52]. The anchor-free head has since been adopted across the YOLO family from YOLOv8 onwards, contributing to improved generalization across object scales [41].

5.1.4. Transformer-Based Detection Methods

DETR (DEtection TRansformer) eliminated hand-designed components—anchor generation and NMS—by casting detection as a set-prediction problem solved through a bipartite Hungarian-matching loss over a fixed pool of learned object queries [53]. A CNN backbone extracts image features enriched with positional encodings; a transformer encoder–decoder then cross-attends these features against N learnable queries, yielding class-label and bounding-box predictions without post-processing. On MS COCO, DETR reaches accuracy comparable to Faster R-CNN, but exhibits weaker small-object performance and notoriously slow convergence—roughly $10 \times$ more training epochs than equivalent CNN-based systems [53].

Deformable DETR addressed both shortcomings by replacing global self-attention with a deformable attention module that samples a small, fixed number of key points around each reference location. This reduces training cost by an order of magnitude and markedly improves small-object recall [54]. In UAV-based surveillance, the RT-DETR-X variant demonstrated that transformer detectors can now operate in real time while exceeding the accuracy of one-stage CNN baselines on small-object benchmarks [52]. Whether transformer detectors will displace one-stage architectures in monitoring applications depends on continued efficiency improvements; the gap in convergence cost and memory footprint relative to YOLO-class detectors remains significant at the time of writing.

5.2. Multi-Object Tracking Methods

Multi-object tracking (MOT) estimates the trajectories of all objects of interest across video frames, assigning consistent identities over time. Three paradigms currently dominate the literature: tracking-by-detection (TBD), joint detection and tracking (JDT), and transformer-enhanced end-to-end approaches [55–57].

5.2.1. Tracking-by-Detection

Architecture. TBD decouples detection and association: a frame-level detector runs independently of the tracking module, and a data-association algorithm links detections across time into consistent trajectories. SORT (Simple Online and Realtime Tracking) established a lightweight TBD baseline using Kalman filtering for motion prediction and

the Hungarian algorithm for optimal assignment based on intersection-over-union (IoU) similarity [58]. DeepSORT augmented SORT with a compact re-identification (Re-ID) network that produces appearance embeddings from detection crops, applying a cascaded matching strategy that resolves recently seen tracks before older ones [59]. ByteTrack reformulated the association step by retaining low-confidence detections rather than discarding them: a first matching round assigns high-score boxes to existing tracklets; unmatched tracklets are then re-associated with low-score boxes, recovering occluded objects that standard thresholding would lose [58]. A recent hybrid approach, SparseDeTrack [60], demonstrates that the TBD framework can accommodate sparse transformer detectors without sacrificing real-time performance. The method retains only 30% of encoder tokens at each attention layer, substantially reducing memory bandwidth while preserving detection quality; the Re-ID module is replaced by an Extended Kalman Filter that directly predicts bounding-box width and height instead of aspect ratio, improving non-linear motion modeling. ByteTrack's two-round association is integrated as the secondary matching stage. On the MOT17 test set, SparseDeTrack achieves 75.4 MOTA at 44.5 FPS, outperforming MOTR, TrackFormer, and TransTrack by 2.0, 1.3, and 0.2 MOTA points, respectively; it also reports 65.6 MOTA on the crowded MOT20 split and 89.1 MOTA on DanceTrack [60].

Performance characteristics. ByteTrack sets a strong TBD reference on the MOT17 and MOT20 benchmarks (see Table 3), and its two-round association remains one of the most practically effective strategies in crowded pedestrian scenes. DeepSORT's Re-ID cascade reduces identity switches significantly over pure IoU matching, at the cost of an additional embedding network and modestly higher latency.

Limitations. The fundamental constraint of TBD is the hard dependency on upstream detection quality. When the detector fails—due to occlusion, motion blur, or low resolution—the association module has no mechanism to compensate, imposing a performance ceiling that JDT and transformer approaches attempt to overcome [57]. ByteTrack's low-score association is a targeted heuristic for this; a general learned solution remains an open problem.

5.2.2. Joint Detection and Tracking

Architecture. The JDT paradigm conducts detection and Re-ID embedding within a shared network, reducing computational redundancy and allowing the two tasks to reinforce each other. CenterTrack conditions the CenterNet detector on two consecutive frames together with a heatmap encoding prior tracklet locations, predicting per-object displacement offsets to the previous frame. This formulation removes the need for explicit feature matching, operating as a lightweight optical-flow surrogate [51]. Chained-Tracker takes a complementary approach: it chains paired attentive bounding-box estimates across overlapping frame pairs through an end-to-end regression architecture [61]. This formulation recasts data association as a direct regression problem, avoiding combinatorial optimization altogether. FairMOT and JDE demonstrated concurrently that learning detection and Re-ID in a one-stage framework can match dedicated TBD pipelines while remaining efficient enough for real-time deployment [56].

Performance characteristics. JDT methods generally reduce missed detections and false positives relative to TBD by sharing features across tasks, and CenterTrack in particular achieves competitive MOTA scores on both MOT17 and the KITTI car category (Table 3) at 22 FPS—a throughput advantage over two-stage TBD systems [51].

Limitations. Performance can saturate in scenes with severe occlusion or rapid motion blur, where the displacement-offset head produces ambiguous predictions [55]. Coupling detection and tracking in a single network also makes the system harder to upgrade incrementally, since improvements to either sub-task may require retraining the full model.

5.2.3. Transformer-Enhanced MOT

Architecture. Transformer-based trackers extend DETR's end-to-end formulation into the temporal domain by maintaining persistent track queries across frames. TransTrack was the first system to exploit this idea: object features from frame $t - 1$ serve directly as track queries for frame t , while a separate set of learnable object queries handles newly appearing targets [62]. MOTR advanced this design by introducing a tracklet-aware label assignment and a Temporal Aggregation Network (TAN) that propagates track query state frame-by-frame without any post-processing NMS or IoU matching, enabling genuinely end-to-end temporal modeling [63]. Global Tracking Transformer (GTR) extended the scope further, jointly modeling appearance and motion over long temporal windows within a single backbone, with demonstrated robustness on large-vocabulary tracking benchmarks beyond pedestrians [64].

Performance characteristics. Transformer trackers excel on benchmarks that stress the association component rather than detection recall. On DanceTrack, which features uniform pedestrian appearance but highly non-linear motion, MOTR surpasses ByteTrack by 6.8 percentage points on HOTA [63]—a striking advantage in a domain where IoU-based assignment breaks down. SparseDeTrack [60] narrows this gap by combining a sparse transformer encoder with a lightweight Kalman-based association, reaching 75.4 MOTA on MOT17 at real-time speed—a result that positions efficient sparse attention as a practical middle ground between full DETR complexity and classical CNN-based TBD. GTR's long-range temporal modeling similarly benefits TAO-style long-tail settings [55,64].

Limitations. On standard pedestrian benchmarks such as MOT17, transformer trackers consistently trail top TBD systems in MOTA. The likely cause is an architectural tension: when detection and tracking share a single decoder, the joint assignment can suppress recall for newly appearing objects, since track queries compete with object queries for attention capacity [63]. Slow convergence and high GPU memory requirements relative to one-stage TBD remain practical barriers to deployment in resource-constrained monitoring hardware [57].

5.3. SOT-Based MOT Methods

A third line of approaches treats each tracked identity as an independent single-object tracking (SOT) problem and aggregates results at the MOT level [56,65]. SOT-based MOT initializes one Siamese or correlation-filter tracker per detected object and propagates each template independently, offering robustness to appearance variation and short-term occlusion because the tracker is dedicated entirely to a single target [65]. Siamese networks embed a template patch and a search region into a shared feature space; because network weights are shared across all instances, the computation scales to multiple tracks without retraining [56]. CenterTrack may be located at the boundary between SOT-inspired and JDT designs: its per-object displacement offsets operate locally on each track rather than globally across all identities, achieving strong local-association accuracy at low inference cost [51]. The primary drawback of full SOT-based MOT is that inference cost grows linearly with the number of simultaneously tracked objects, making it ill-suited to dense crowd monitoring. A second weakness is sensitivity to initialization errors: without a global consistency mechanism, a mis-initialized template can contaminate an entire trajectory [57,65].

In 3D LiDAR-based tracking, Yin et al. [66] combined a pillar-based CenterPoint detector with a velocity-conditioned Kalman filter, demonstrating that SOT-style per-object state propagation scales effectively to autonomous driving pipelines evaluated on the Waymo Open Dataset [67]. The approach is notable because it operates on point clouds rather than images, highlighting that the SOT-based paradigm is not inherently tied to 2D appearance features. Xu and Huang [68] extended SOT-style per-object propagation to the

multi-agent collaborative setting with CoTrack. The method builds on a spatio-temporal aggregation module comprising two complementary sub-modules: the spatial cross-agent collaboration sub-module dynamically integrates feature maps from neighbouring vehicles to correct localization errors, while the temporal ego-agent enhancement sub-module maintains historical frame context to compensate for data sparsity in sparse point-cloud observations. An unsupervised feature compressor reduces inter-agent communication bandwidth, and a two-stage online association aligns detection-track pairs across agents. Experiments on both simulated and real datasets confirm state-of-the-art performance in collaborative 3D detection and tracking under noisy, bandwidth-limited conditions [68]. CoTrack thus extends the theme established by CenterPoint [66]—that SOT-style temporal propagation transfers well to point-cloud pipelines—to the multi-agent, vehicle-to-vehicle perception context.

Table 3. Summary of representative object detection and multi-object tracking methods.

Method	Category	Benchmark	Reported Accuracy	Speed/Complexity
SSD [45]	Det/One-stage	VOC2007	74.3% mAP ^a	Single-shot CNN detector
YOLOv4 [42]	Det/One-stage	COCO	43.5% AP; 65.7% AP50	62 FPS (V100); 33 FPS (Pascal)
YOLOv7-E6E [46]	Det/One-stage	COCO	56.8% AP ^b ; 74.4% AP50	151.7 M params; 843.2 GFLOPs; 36 FPS (V100)
YOLOv8 [47]	Det/Anchor-free	COCO	53.9% AP	Anchor-free YOLO head; speed/complexity NR
DETR [53]	Det/Transformer	COCO	42.0% AP; 62.4% AP50	41 M params; 86 GFLOPs; 28 FPS; 500 epochs
Def-DETR [54]	Det/Transformer	COCO	46.2% AP; 65.2% AP50	40 M params; 173 GFLOPs; 19 FPS; 50 epochs
ByteTrack [58]	MOT/TBD	MOT17	80.3 MOTA; 77.3 IDF1; 63.1 HOTA	29.6 FPS; YOLOX + two-stage association
DeepSORT [59]	MOT/TBD	MOT16	61.4 MOTA; 62.2 IDF1	40 Hz; Re-ID pass ~30 ms/32 boxes
CenterTrack [51]	MOT/JDT	MOT17/KITTI (car)	67.8/89.4 MOTA; 64.7 IDF1	22 FPS (MOT17); 15 FPS (KITTI)
TransTrack [62]	MOT/Transformer	MOT17 test	74.5 MOTA; 63.9 IDF1	10–15 FPS in decoder ablation
MOTR [63]	MOT/Transformer	DanceTrack	54.2 HOTA; 79.7 MOTA; 51.5 IDF1	7.5 FPS (V100)
SparseDeTrack [60]	MOT/TBD + sparse transformer	MOT17/MOT20/ DanceTrack	75.4/65.6/89.1 MOTA	44.5 FPS; 30% token retention
CenterPoint [66]	MOT/JDT	Waymo/nuScenes	71.8/66.4 mAPH ^c ; 63.8 AMOTA	11 FPS (Waymo); 16 FPS (nuScenes)

^a VOC 2007 test set (11-point interpolated AP). ^b MS COCO test-dev, input resolution 1280 × 1280. ^c Waymo Open Dataset test split, LEVEL_2 mAPH for vehicle and pedestrian classes. NR indicates that the corresponding speed or complexity indicator was not reported in the analyzed source.

The values in Table 3 should be interpreted as a structured comparison of reported accuracy, speed, and complexity indicators rather than as a strict ranking. Direct numerical comparison is limited by differences in datasets, input resolutions, hardware platforms, and evaluation protocols across the reviewed studies.

5.4. Benchmarks and Evaluation Metrics

A range of standardized benchmarks has driven progress in both detection and tracking. Table 4 summarizes the principal datasets and their scope.

MOTChallenge. The MOTChallenge suite (MOT15, MOT16, MOT17, and MOT20) provides pedestrian-dominated sequences with ground-truth trajectories for standardized evaluation [69]. MOT17 offers 14 sequences annotated at whole-body level with three public detection sets (DPM, Faster R-CNN, and SDP), enabling controlled comparison of

association algorithms in isolation from detector choice. MOT20 increases crowd density markedly, stressing trackers under occlusion and clutter conditions absent from MOT17. ByteTrack’s published results on both splits remain a widely used reference point for the TBD paradigm [58].

KITTI. KITTI comprises training and test sequences with annotations for up to eight object classes and is the standard benchmark for autonomous-driving detection [43]. Its inclusion of stereo imagery and, in extended versions, point-cloud data supports both 2D and 3D evaluation protocols, making it a useful bridge between purely image-based and LiDAR-based detector comparisons.

Waymo Open Dataset. It provides 1150 driving segments annotated for 2D and 3D bounding boxes across five sensor modalities, covering diverse geographic and weather conditions [67]. It serves as the primary evaluation venue for LiDAR-based detectors including PointPillars [67,70], PointRCNN [67,71], STD [67,72], and CenterPoint [66,67].

BDD100K. BDD100K contains 100K driving videos spanning diverse weather, illumination, and geographic conditions, with a dedicated MOT split of 1400 training, 200 validation, and 400 test videos across eight object classes [73]. The dataset’s scale and diversity make it a valuable complement to the pedestrian-centric MOTChallenge suite.

TAO and DanceTrack. TAO (Tracking Any Object) expands the category scope dramatically to 2907 high-resolution videos spanning 833 object classes, far beyond the category diversity of MOTChallenge, specifically targeting long-tail tracking performance [74]. DanceTrack takes a different approach: by providing sequences with deliberately uniform pedestrian appearance but highly diverse, non-linear motion patterns, it isolates the association component as the primary challenge, making it the most demanding benchmark for evaluating tracking-by-motion rather than tracking-by-appearance [63].

Evaluation Metrics. The CLEAR-MOT suite defines MOTA as a combination of false positives, false negatives, and identity switches normalized by ground-truth count [69]. Because detection failures dominate this combination, MOTA tends to reward strong detectors more than strong association strategies. IDF1 counters this by evaluating identity preservation at trajectory level through bipartite matching that maximizes temporal overlap, providing a complementary view of association quality. HOTA was designed to address the known MOTA bias by explicitly separating detection accuracy, association accuracy, and spatial localization into a single scalar—a property that has made it the preferred metric on DanceTrack and increasingly on MOTChallenge [58].

Table 4. Principal benchmarks for object detection and multi-object tracking.

Benchmark	Year	Domain	Classes	Primary Metric
MOT17 [69]	2017	Pedestrian	1	MOTA, IDF1
MOT20 [69]	2020	Crowded pedestrians	1	MOTA, HOTA
KITTI [43]	2013	Driving	8	AP, MOTA
Waymo [67]	2020	Driving (LiDAR)	4	mAPH, L2
BDD100K [73]	2020	Driving	8	mMOTA
DanceTrack [63]	2022	Dance (uniform ap.)	1	HOTA, AssA
TAO [74]	2020	Long-tail, diverse	833	Track AP

Open challenges and future research directions for detection and tracking are consolidated in Section 7.

6. Specialized Applications

In the area of specialized applications of object detection and movement monitoring, several leading topics have been identified, including action recognition, autonomous driving, and motion analysis & surveillance.

6.1. Autonomous Driving

Controlling an autonomous vehicle requires a fast and error-free system for detecting surrounding objects and monitoring their movement. The problems and challenges posed by such systems using deep learning techniques are presented in papers by Gupta et al. [75], Berwo et al. in [76], and the recent research progress on deep learning-based 3D object detection algorithms by Cai and Wu [77]. Moreover, Haris et al. [78] provided a comprehensive survey of image-based three-dimensional object detection approaches.

Among the recent research papers, Zhang et al. [79] attempted to improve object detection for autonomous driving in snowy conditions. They built a lightweight detection model that explicitly addresses feature occlusion and spatial distortion caused by snowfall. The model integrates three novel modules into a YOLOv8s-based framework, including: Multivariate Feature Learning (MFL) for richer feature extraction under snow occlusion, an Associative Spatial Information (ASI) Enhancement module to strengthen cross-dimensional spatial awareness, and a Lighter Fusion C2f (LF_C2f) mechanism to reduce computational cost while maintaining accuracy. Experimental verification on the real-world RSOD dataset obtained 59.8% mAP@0.50 and 40.7% mAP@0.50–0.95, outperforming multiple state-of-the-art detectors while retaining a lightweight profile (9.7M parameters and 27.4 GFLOPs). The other proposal for object detection under challenging conditions, such as low illumination or adverse weather, was made by Zhang et al. [80]. The designed dual-stream detection system integrates two novel modules: Adaptive Dual Cross-Attention (ADCA), which flexibly balances self-attention and cross-attention between modalities via learnable weights, and Adaptive Max-Average Downsampling (AMAD), which enhances multi-scale feature representation through dynamically weighted max- and average-pooling. The demonstrated performance on individual dataset equals to 86.2% mAP@50 (DVTOD), 97.6% mAP@50 (LLVIP), and 78.9% mAP@50 (FLIR), outperforming previous known methods. An alternative framework for improving multimodal 3D object detection that addresses the noise and inconsistency introduced by image-based depth completion, was presented by Song et al. [81]. The designed DRNS (Dual-branch Noise-Resistant Sparse convolution) framework unifies de-noising, multimodal fusion, and iterative feature refinement. The achieved 3D Car AP (R40) performance on KITTI reached 95.76%/89.69%/87.89% for Easy/Moderate/Hard, respectively.

In turn, Lu et al. [82] created an SDF-YOLO framework based on YOLOv8n for improving the detection of small objects. The created system includes Sim-ACCoM modules for multi-dilation convolutions and attention for contextual and cross-scale feature enhancement, Dynamic Interpolation Fusion (DIF) modules for efficient same-scale feature alignment and fusion, and Group-Slim pruning to reduce parameter count while maintaining accuracy. The experimental results on the KITTI and BDD100K datasets show improvements: mAP of +3.3% to +3.7%, Recall of +3.4%, and Precision of +1.7% compared to YOLOv8n. The achieved Precision was 79.5% mAP@50 and 57.1% mAP@50–95, while model size was reduced by 1.2 MB. Moreover, Foroutannia et al. enhanced the accuracy and robustness of 3D object detection for autonomous driving by fusing camera and LiDAR data [83]. The proposed system first generates 2D bounding boxes from camera images using YOLOv4, then projects them into LiDAR space using a geometric frustum method to extract relevant 3D point clouds, and finally applies a 3D U-Net architecture for precise 3D bounding box segmentation. The experimental results on the KITTI dataset show high detection performance, up to 96.96% global accuracy and 91.06% weighted IoU (0% noise), maintaining 83.48% accuracy and 82.76% weighted IoU at 10% noise. The next proposal for improving small-object detection accuracy in complex autonomous driving environments using LMSDF-Net was delivered by Wu et al. [84]. The YOLOv11-based lightweight framework (2.4 M parameters and 9.9 GFLOPs) was enhanced with a hybrid attention module to suppress background noise, a multi-scale dynamic feature-

focusing module to better aggregate discriminative features, and a P2 small-object detection head. The experimental results on Udacity and SODA10M datasets showed 86.2% mAP@50 and 56.3% mAP@50–95 (Udacity), 63.6% mAP@50 and 44.0% mAP@50–95 (SODA10M). Moreover, a geometry-guided BEV detection framework for improving camera-based multi-view BEV 3D object detection was proposed by Pan et al. [85]. The proposed system includes a Deformable Multi-Task (DMT) head, a Normal-To-Depth (N2D) module, a parameter-free Local Planar Depth Aggregation (LPDA) module, and a normal-aware BEV representation forming module. The nuScenes benchmark results show 61.9% NDS and 52.8% mAP on the test set, outperforming equivalent BEV detectors with minimal additional computational cost. The VoteFusion framework for improving robust 3D object detection under occlusion and noise, which combines radar/LiDAR point clouds with multi-view image features, was developed by Wang et al. [86]. The framework introduces a novel voting-based cross-modal verification mechanism to generate high-confidence 3D proposals. The experimental results on KITTI, nuScenes, and Waymo demonstrate strong performance: 90.30% mAP (BV), 84.33% mAP (3D) on KITTI, 72.4% mAP and 75.4% NDS on nuScenes, and 75.1/73.5 mAP/mAPH on Waymo. A next proposal, intended particularly to support the detection of small and occluded objects in complex traffic scenes, was delivered by Liu et al. [87]. For this purpose, an enhanced RetinaNet-based detector, AttenRetina, with a Feature Pyramid Network (FPN) and a hybrid spatial-channel attention module was designed. The experimental results include mAP = 0.86 (KITTI) and mAP = 0.82 (MS COCO), with Precision/Recall of 0.92/0.89 (KITTI), improving the base RetinaNet mAP by over 12%. Similarly, a robust, real-time 3D detection framework for small and distant objects, consisting of a parallel-connected 2D RGB object detector and a 3D LiDAR detector, was designed by Sgaravatti et al. [88]. Experiments on the KITTI dataset showed a noticeable improvement in detecting imbalanced object classes, such as pedestrians and cyclists, compared with other LiDAR-only and multimodal methods. The main features and parameters of the most recent methods designed for autonomous car object detection are summarized in the Table 5.

Table 5. Comparison of object detection methods for autonomous vehicles.

Type	Pub.& Method	Sensors	Datasets	Key Metrics	Main Contribution
2D	[79] MAL-Det (2026)	RGB Cam.	RSOD, BDD100K	mAP@0.50 = 59.8% mAP@0.50–0.95 = 40.7%	Lightweight, for snow object detection, YOLOv8s-based
	[82] SDF YOLO (2026)	RGB Cam.	KITTI, BDD100K	mAP50 = 79.5% mAP50–95 = 57.1%	Lightweight, dynamic multi-scale feature fusion, controlled parameter reduction
	[84] LMSDF Net (2026)	RGB Cam.	Udacity, SODA10M	Ud.: mAP@0.5 = 86.2% mAP@0.5–0.95 = 56.3%	Hybrid spatial-channel attention
	[87] AttenRetina (2025)	RGB Cam.	KITTI, MS COCO	mAP = 0.86 (KITTI), 0.82 (MS COCO)	dynamic multi-scale feature fusion RetinaNet + FPN + spatial/channel attention
3D	[85] Geo-G.BEV (2026)	RGB multiview	nuScenes	test:NDS = 61.9%, mAP = 52.8%	Using surface normals for depth estimation and BEV construction
Multimod. 2D	[80] ADCA-Net (2026)	RGB Cam., Thermal	DVTOD, LLVIP	mAP@50 = 86.2% DVTOD, 97.6% LLVIP	Adaptive Dual Cross-Attention for flexible RGB–thermal fusion
Multimod. 3D	[83] 3D U-Net env. (2026)	LiDAR + RGB Cam.	KITTI	3DUNet global acc. train: 96.9%, test: 83.5%	Two-step 3D detect.: cam.–LiDAR + 3DUNet volum. segmentation
	[86] VoteFusion (2026)	LiDAR + Cam. + radar	KITTI, nuSc.	KITTI: mAP_3D 84.33%, mAP_BEV 90.30%	Object Attention for Efficient LiDAR–Image Fusion in 3D Detect.
	[88] LCF3D (2026)	LiDAR + RGB Cam.	KITTI, nuScenes	NDS ≈ 0.674 nuScenes: mAP ≈ 0.635,	Hybrid cascade RGB–LiDAR fusion, improved domain generalization
	[81] DRNS (2026)	LiDAR + RGB Cam.	KITTI	3D AP 95.8/89.7/87.9 (Easy/Mod./Hard)	Noise-resist. dual-branch LiDAR-camera fusion with sparse convol.
	[89] SAMUNet (2025)	LiDAR	Waymo, nuScenes	nuSc.: 67.7% mAP, 72.0% NDS	Sparse Mini-Unet + DMU + Shape-aware Center Head

Table 5 shows that most recently created models use 2D methods with only RGB cameras or multimodal 3D methods that combine cameras and LiDAR components. The most commonly used public datasets include KITTI, nuScenes, BDD100K, Waymo and several others. Object detection quality metrics usually include classic mAP (mean average precision). Moreover, some dataset-specific metrics, such as NDS (nuScenes Detection Score), are also utilized.

6.2. Vision-Based Human Activity Analysis and Surveillance

The following subsections analyze AI-assisted human action recognition and surveillance systems, respectively.

6.2.1. Action Recognition

An enhanced survey detailing video action recognition (VAR) systems was performed by Gong and Luo [90]. The authors analyzed almost one hundred publications concerning RGB-based or HPE-based models and distinguished main architectures of the models: two-stream networks, 3D or Graph Convolutional Neural Networks (3D-CNNs/GCNs), recurrent neural networks or Long Short-Term Memory Networks (RNNs/LSTMs), and Attention Mechanism (AM). An extensive analysis of over 200 publications was conducted by Zhu et al. [91]. In this case, the authors reached similar conclusions, indicating that initial solutions relied on adaptive deep learning, while currently three technologies are leading: two-stream networks, 3D convolutional kernels, and the most recent compute-efficient models. A similar analysis, although narrower in scope, was provided by Tang et al. [92]. The authors' conclusions largely align with earlier ones and also point to the growing role of transformer-based methods that combine natural language processing with computer vision. The important role of deep learning techniques in human action recognition using artificial neural networks such as CNN, RNN, LSTM, DBN (Deep Belief Network), and others was also emphasized by Pham et al. [93] based on over 230 references from the 90s of the XX century to 2020.

Concerning the action recognition related to the human bodies, a video-based pedestrian attribute recognition (PAR) system was built by Wang et al. [94]. To overcome the limitations of image-based methods and inefficient video models, the authors created a vision-language VTFFPAR++ framework that leverages a frozen CLIP foundation model together with a novel spatio-temporal side-tuning strategy for parameter-efficient adaptation. The experimental results on the MARS and DukeMTMC-VID datasets show reaching the 93.1% accuracy and 83.1% F1-score (MARS), and 93.3% accuracy and 80.2% F1-score (DukeMTMC-VID), with far fewer trainable parameters and lower memory demands than other existing methods. In turn, Shin et al. [95] proposed a video-based human activity recognition (HAR) system, which combines a pre-trained ConvNeXt model for per-frame spatial feature extraction with a Temporal Convolutional Network (TCN) for sequence modeling. Experiments on four benchmark datasets demonstrate high performance, achieving 97.7% accuracy (UCF11), 98.8% (UCF50), 98.5% (UCF101), and 83.4% (JHMDB). The other existing LSTM- and transformer-based approaches were outperformed by up to 7%, providing faster inference and lower memory demand. Another proposal for a multi-pose HAR system from UAV-captured video was presented by Sinha & Kumar [96]. The system is based on a novel DMLC-CNN framework, and can compensate for the influence of perspective distortion, orientation errors, and complex backgrounds. The experimental results showed that the model outperformed standard CNN, DNN, DBN, and RNN baselines, achieving 94.5% accuracy, lower false-positive/negative rates (~ 0.044), reduced clustering time (36 ms), and significantly shorter training time. Moreover, Feichtenhofer et al. [97] proposed a method for improving video action recognition and detection by explicitly

modeling the different roles of spatial semantics and temporal motion. The developed SlowFast network architecture combines two pathways of video signal processing: a Slow pathway, sampling video sparsely to capture rich spatial semantics, and a Fast pathway, sampling densely with fewer channels to capture fine-grained motion. The experimental results show excellent performance across multiple benchmarks, including 79.8% top-1 accuracy (Kinetics-400), 81.8% top-1 (Kinetics-600), 45.2% mAP (Charades), and 30.7% mAP (AVA action detection).

6.2.2. Intelligent Surveillance and Anomaly Detection

Another specialized application of dynamic object detection techniques is motion analysis and surveillance. Publications in this field typically include methods using convolutional neural networks with deep learning or transformer-based techniques. A survey comparing both mentioned attempts was provided by Arkin et al. [98]. A more specific analysis of pedestrian detection and tracking was presented by Brunetti et al. [99].

Sun et al. [100] attempted to improve unsupervised video anomaly detection for surveillance by proposing SMACAN, a hyperparameter-free, scene- and motion-aware anomaly detection framework. The designed framework has a dual-branch auto-encoder architecture (frame prediction + optical flow reconstruction), enhanced with a Scene-Object Refinement Module (motion calibration), a Cross-Flow Attention Module (motion-appearance fusion), and a Motion Feature Distribution Module (anomaly detection). The experimental results show high performance, achieving 83.1% (ShanghaiTech) and 92.5% (CUHK Avenue), outperforming other reconstruction-, prediction-, and hybrid-based methods. In turn, Wang et al. [101] built an integrated multi-object tracking pipeline that jointly addresses low-SNR detection, long-term motion modeling, and binocular 3D trajectory reconstruction. The proposed system combines a two-stage cascade detector (global enhancement + local noise suppression), a hybrid tracker (Kalman filter + Transformer-based motion model), and binocular matching (reconstruction of 3D trajectories). The simulation results show improvements over baselines, achieving 98.0 MOTA, 92.7 IDF1 (2D tracking), and 98.4 MOTA and 94.0 IDF1 (3D tracking). Hu et al. [102] developed a four-layer (retina layer, SGC dendritic layer, SGC soma layer, and a two-stage Rt layer), biologically inspired architecture of a small moving object detector in complex, high-altitude scenes. Experiments on the BEVS and RIST datasets showed higher detection rates and lower false alarm rates than classical methods (e.g., frame difference, optical flow, and Gaussian models). Furthermore, Ness's work [103] concerns improving vehicle detection and recognition in smart surveillance systems. The designed comparative LSTM-based framework uses motion direction and motion angle features to distinguish vehicles from other objects in video data. The experimental results show the accuracies of 81% and 83% on UCSD and VRiV datasets, respectively. A similar xLSTM-based lightweight tracker for robust and efficient video-level visual object tracking was proposed by Jia et al. [104]. The system's architecture comprises a frame-level motion-aware module and a video-level motion-aware module that uses an xLSTM-based recurrent implicit motion state to integrate long-term temporal dynamics. The tracker was tested using eight benchmarks. Example result on LaSOT equals to 75.2% AUC at 55 FPS with only 69.6 M parameters and 44 GFLOPs. The next paper by Rajasekaran et al. [105] aims to improve multi-target object detection and classification for smart IoT surveillance. The proposed system consists of a noise removal module, a semantic feature extraction module, a Multi-Component Attention Graph Convolutional Neural Network for detection and classification, and a Brown Bear Algorithm to optimize network weights. Experiments on the PASCAL VOC dataset achieve 99.5% accuracy, 99.1% specificity, and 99.1% mAP, maintaining low model size (60.80 MB) and real-time speed (53 FPS). In turn, Jiang et al. [106]

attempted to reduce identity switches in multi-object tracking under occlusion by modeling motion in the 3D world coordinate system instead of the 2D image plane. A novel tracker integrating a linear camera model with a dimension recovery module, a world motion model (WMM) and a combined 2D-3D cost matrix was proposed. The experimental results show improved identity preservation, e.g., IDF1 = 81.2, HOTA = 65.7, MOTA = 80.8 (MOT17 dataset), maintaining real-time performance comparable to motion-only trackers. Finally, Kumar et al. [107] proposed improvements to object detection for UAV-based smart surveillance. A YOLOv5-based deep neural network, enhanced with three multi-scale prediction heads, was designed to address small object sizes, dense scenes, and occlusion in aerial imagery. The experiment results using the VisDrone dataset show 95.1% precision, 86.3% recall, and 91.6% mAP@0.5, outperforming standard YOLOv5 and YOLOv7 models. Descriptions of the designed models, evaluation methods, outcomes, as well as main contributions of the most recent works in this area are summarized in Table 6.

Table 6. Comparison of motion analysis and surveillance methods.

Purpose	Pub.& Method	Model	Datasets	Evaluation	Main Contribution
Object tracking	[101] KF + Trans (2025)	Transformer + Kalman filter	STKEngine generated	2D:MOTA = 98.0, IDF = 92.7	Method for multi-object tracking in deep space considering low SNR
	[104] IMSETrack (2026)	Recurrent ANN	LaSOT, GOT and other	3D: MOTA = 98.4, IDF = 94.0 LaSOT: 75%, GOT-10k: 77.7%	Video-level object tracking method
	[106] WM-SORT (2025)	Transf. + xLSTM Analytic 2D-3D model	MOT17 & 20 DanceTrack	TrackingNet: 86.8%; MOT17: IDF1 = 81, MOTA = 81 MOT20: IDF1 = 78, MOTA = 76	Multi-object tracking method in the 3D domain
Object detection	[105] MTO-IoT MCAGCNN (2025)	CNN-based GCNN + attention	PASCAL VOC 2012	Accur.: 99.5%, Specif: 99.1%	Multi-target detection method in IoT
	[107] Enhanced YOLOv5 (2026)	YOLOv5	VisDrone, MS COCO	mAP: 99.1%, Precision: 99.2% Precision: 95.1–95.4%, mAP@0.5: 91.6–95.3%	Method for object detection in UAV images
Object recognition	[102] TSOM (2025)	Bio-inspired architecture	BEVS, RIST	Accuracy: from 0.33 to 0.75–0.80	Bio-inspired method for detection & recogn. of small movable objects
	[103] Smart Surv. System (2025)	Recurrent ANN LSTM	UCSD, VRiV	Accuracy: 81% (VRiV), 83% (UCSD)	Method for vehicle detection and recognition
Anomaly detection	[100] SMACAN (2025)	Autoencoder unsupervised	CUHK Ave. ShanghaiTech	AUC: CUHK = 92.5%, ShanghaiTech = 83.1%	Method for anomaly detection in monitoring systems

Most of the developed frameworks focus on 2D space; only two models [101,106] operate in 3D. The most commonly used quality measures are detection metrics (mAP) and tracking metrics (MOTA, IDF1). In isolated cases, the anomaly metric AUC [100] and the biological-perceptual metrics DR and FA [102] are used. The built models, except [106], are based on various artificial neural network (ANN) architectures. Supervised training currently dominates among ANN tuning methods, although hybrid [101], multi-set [104], and unsupervised [100] approaches also appear.

7. Future Research Directions and Challenges

This section consolidates the main future research directions and open challenges identified across the reviewed streams.

7.1. Human Pose Estimation (HPE)

- Looking ahead, in medical diagnostics examinations, low-latency pose or motion monitoring in the scanner workspace can support objective assessment of patient movement and help mitigate motion-related artefacts through timely feedback or scan control. Future work should prioritize robustness under constrained viewpoints and occlusions, lightweight on-device inference, and uncertainty-aware outputs suitable for safety-critical workflows.
- From a technology and architecture perspective, HPE research is expected to continue toward hybrid, hardware-aware models that combine transformer-style global context

with efficient local inductive biases (e.g., convolutions, graph layers, or state-space blocks). A key direction will be to support longer temporal windows and better use of information about body structure without prohibitive attention cost, while co-designing training recipes and inference optimization for predictable latency on deployment hardware.

- Multimodal HPE is expected to grow in practical importance, combining RGB with complementary sensing (e.g., depth, IMU/RF, and point clouds) and exploiting multi-view geometry when available. The central challenge is robust sensor fusion under occlusion and constrained viewpoints, including graceful degradation when modalities are missing, noisy, or mis-calibrated. Multimodal designs can also enable privacy-preserving deployments by reducing reliance on identifiable RGB imagery while maintaining clinically meaningful motion assessment.

7.2. Object Detection and Tracking

Several fundamental problems remain unresolved across detection and tracking.

- Small-object detection is particularly acute in UAV and wide-area surveillance, where targets occupy only a few pixels; slicing-based inference pipelines such as SAHI offer partial relief but impose throughput penalties that preclude real-time use [52].
- Occlusion continues to erode both detection recall and trajectory coherence in densely crowded scenes; ByteTrack's two-round low-score association is a pragmatic heuristic for this, but the broader problem of reasoning through mutual occlusions has not been resolved in the literature [57,58]. Current mitigations: ByteTrack's two-round association, MOTR's collective average loss (CAL), and attention-based re-identification, remain task-specific heuristics rather than general solutions [58,63].
- Computational efficiency is critical for edge deployment; the accuracy advantages of transformer detectors (DETR and Deformable DETR) currently come with slow convergence and high memory footprint that limit their use on embedded monitoring hardware [53,54]. The efficiency constraint is particularly acute for embedded deployment. Hardware-software co-design on FPGA platforms (e.g., Vitis AI with YOLOv3 on Xilinx ZCU104) demonstrates a viable path to inference without GPU-class resources: the DPU-accelerated detection module achieves 21.4 FPS system throughput ($1.53\times$ faster than a CPU-only baseline), while the hybrid tracking system attains an 81% success rate at the corresponding similarity threshold [108]. Nevertheless, the accuracy gap relative to server-grade GPU pipelines remains non-negligible, and lightweight transformer architectures capable of closing that gap are still an open research problem [60].
- Domain shift between benchmark training distributions and real monitoring environments—varying weather, night-time scenes, wide-area aerial perspectives—limits the transferability of models trained on MOT17 or COCO [73,109].
- Long-tail category coverage is targeted by TAO, yet no current tracker generalizes robustly across its 833 categories at levels approaching single-category pedestrian performance [74].
- Temporal consistency over long sequences remains a weakness of transformer trackers; MOTR's collective loss and TAN improve inter-frame continuity, yet the architecture still trails TBD baselines on standard pedestrian benchmarks [63].

Productive future directions include lightweight transformer architectures that pair deformable attention with efficient backbones, unified end-to-end detection–tracking models exploiting point-cloud and image fusion for improved 3D situational awareness [66], and self-supervised pre-training strategies capable of reducing annotation dependency across the diverse conditions of real-world monitoring deployments [55,57].

8. Conclusions

Over the past few years, computer-aided object position and movement detection systems have become a topic of research in a significantly growing number of research centres. Due to their specific nature, these systems are typically built based on artificial intelligence technologies, especially various neural network architectures. This study focuses on scientific articles mainly published since 2019, available in the Scopus and Web of Science databases. A preliminary analysis of the topic identified three main areas of research, including human pose estimation, object detection and tracking, and specialized applications. Subsequently, more detailed analyses were conducted within each area. Finally, conclusions of a scientific, practical, and future-oriented nature are presented. Based on the analysis, the following generalized conclusions can be drawn:

- Within the human pose estimation (HPE) stream, the reviewed literature highlights a clear progression from neural taxonomies and benchmarks toward transformer-based backbones and efficient real-time pipelines that increasingly make both 2D and 3D pose recovery practical in operational settings.
- The YOLO family of one-stage detectors—from YOLOv4 through YOLO11—has consistently advanced the speed–accuracy frontier on standard benchmarks such as MS COCO, with each generation introducing architectural refinements (CSP connections in YOLOv4, anchor-free C2f/C3k2 heads in YOLOv8–YOLO11, depthwise convolutions in the YOLO11 detection block) that improve the parameter-to-accuracy ratio without sacrificing real-time throughput [41,42,46,47]. These properties make YOLO-class detectors the preferred backbone for practical monitoring deployments. Transformer-based detectors (DETR and Deformable DETR) offer complementary gains in set-prediction quality, but at the cost of significantly higher convergence time and memory footprint [53,54].
- In multi-object tracking, no single paradigm dominates across all evaluation conditions. Tracking-by-detection methods such as ByteTrack achieve state-of-the-art MOTA on crowded pedestrian benchmarks (MOT17: 80.3 MOTA, 30 FPS; MOT20: 75.2 MOTA) through two-round low-score association [58]. By contrast, transformer-enhanced end-to-end trackers such as MOTR surpass ByteTrack on association quality on DanceTrack, where appearance is uniform and motion is non-linear, recording a 6.8 percentage-point gain in HOTA and an 8.1 percentage-point gain in AssA, because IoU-based assignment breaks down in that regime [63]. Sparse-attention hybrid approaches such as SparseDeTrack [60] represent a productive middle ground: within the Sparse DETR detection stage, a 30% token-retention rate reduces encoder FLOPs by 14% relative to the 50% retention configuration (169.48 G vs. 197.09 G FLOPs in the token-ablation study), while the full system maintains a real-time inference speed of 44.5 FPS. SparseDeTrack thereby ranks first among transformer-based methods on MOT17 (MOTA 75.4), suggesting that selective attention compression is a viable design direction for resource-constrained deployments.
- 3D object detection and tracking, whether from point clouds or cooperative multi-agent sensor fusion, has matured sufficiently to be evaluated on large-scale benchmarks for autonomous driving (Waymo, KITTI, and nuScenes), with CenterPoint combining pillar- and voxel-based detection with greedy, velocity-aware nearest-neighbour association to establish a strong 3D tracking baseline (63.8 AMOTA on nuScenes, outperforming the prior state-of-the-art by 8.8 AMOTA) [66]. Recent cooperative extensions such as CoTrack [68] demonstrate that spatio-temporal feature sharing across multiple agents can compensate for the localization errors and data sparsity inherent in vehicle-to-vehicle perception. CoTrack employs deformable cross-attention for spatial alignment (Spatial Cross-agent Collaboration module) and criss-cross atten-

tion for temporal context (Temporal Ego-agent Enhancement module). These results indicate that the paradigmatic separation between single-sensor and multi-sensor tracking is narrowing. Consolidating these 3D pipelines with image-based 2D methods into unified detection-tracking architectures remains an important direction for future research [66,68].

- Specific practical applications span a wide range of fields, but particularly numerous studies focus on issues such as action recognition, object and movement detection, autonomous driving, motion analysis, and surveillance.
- Among these specific applications, until a few years ago, 2D imaging techniques were widely used. Currently, multi-object 3D systems based on convolutional neural networks (CNNs), recurrent neural networks (RNNs) or transformers are becoming increasingly common. Although these systems have higher hardware requirements, they offer greater accuracy, especially in noise or low-quality signal environments.

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